

Arithmetic Power Geometry III: The Integrated Closure Defect Functional and the Conditional Frey-Curve Height Obstruction Program

Mohammad Amir Khusru Akhtar
Usha Martin University, Ranchi – 834001, Jharkhand, India
akakhtar.2024@gmail.com

Abstract

We establish an exponent-globalized analytical framework for Arithmetic Power Geometry (APG) by constructing a path-integrated analytical system over the continuous exponent deformation space. Moving beyond localized static evaluations and artificially regularized tensor fields, this paper introduces the Integrated Closure Defect Functional (A_0), defined as the unregularized path integral of the exact continuous closure defect from the classical Euclidean baseline ($n = 2$) to an arbitrary deformed exponent state ($n > 2$). We state and prove two foundational internal theorems: the strict positivity of the integrated closure defect functional for arbitrary positive base coordinates, and its third-order asymptotic expansion near the baseline, governed explicitly by the normalized coordinate Shannon entropy ($H(W)$) and a newly isolated quadratic logarithmic weight dispersion parameter ($Q(a, b)$). Finally, we formalize a rigorous, conditional Frey-Curve Height Obstruction Program. We detail the proposed cross-disciplinary criteria under which this continuous exponent-globalized path integral can be coupled to the discrete arithmetic attributes of semi-stable elliptic schemes, defining the precise target bridge theorems required to interface continuous geometric defect functionals with the modular representation landscape.

Keywords: Arithmetic geometry; exponent deformation; information geometry; Shannon entropy; Faltings height; Frey curve; modular forms.

1 Introduction and Architectural Exponent-Globalization

Classical investigations in arithmetic geometry and Diophantine analysis typically focus on the structural and geometric properties of rational or integral solution sets lying on algebraic varieties defined by polynomial equations with immutable algebraic exponents. In that structural paradigm, the coordinate fields are treated as variables evaluated over fixed arithmetic rings and fields.

Arithmetic Power Geometry (APG) introduces an inverted structural perspective. It stabilizes the positive integer base coordinates and treats the equation's power exponent as a continuously varying parameter over a specialized differentiable parameter space, the Power-Deformation Space. The geometric viewpoint adopted here is conceptually influenced by classical differential geometry, where parameterized spaces are analyzed through local coordinate structures, metric properties, and curvature behavior [11, 18]. The foundational motivation for this perspective arises from the deep qualitative threshold separating the classical Pythagorean baseline ($n = 2$), where algebraic closure is abundant over infinite discrete subsets, from higher-power regimes ($n > 2$), where exact integer closure is restricted by the deep properties of Galois representations and modular forms.

These arithmetic obstructions are closely related to modern Diophantine geometry, height theory, and the arithmetic of algebraic varieties [5, 16, 28, 14].

In APG I, a localized functional tear—the Local Closure Defect (δ_K)—was evaluated statically at the unique baseline coordinate designated as the Euclidean Target. APG II extended this perspective by introducing a regularized information-geometric tensor field to analyze the continuous parametric sensitivity of the manifold, drawing structural methodologies from classical information geometry and differential geometric potentials.

However, both foundational treatments remained strictly local with respect to the exponent variable, characterizing the instantaneous structural strain at or near the boundary $n = 2$. As noted in adjacent literature on Diophantine approximation landscapes, local continuous metrics are frequently blind to global number-theoretic obstructions due to dense, non-isolated near-miss phenomena.

This paper establishes the exponent-globalization of APG. Rather than measuring the static geometric friction at a solitary parameter coordinate, we integrate the continuous deformation sequence across the entire path stretching from the Euclidean baseline to a higher-power target. To preserve the algebraic purity of the framework, we abandon the diagonal regularizers (λ) employed in APG II, constructing a pure, unregularized path-dependent analytical framework. We present the mathematical foundations of this integrated closure defect, prove its core local asymptotic growth profiles, and formalize the conditional mathematical program linking continuous analytical defects to the discrete arithmetic heights of the Frey curve representation space.

2 The Integrated Closure Defect Functional (A_0)

To ensure that this path-globalized analytical framework represents an intrinsic property of the underlying Diophantine parameters, we define the path integral using only the exact physical closure defect.

Let $(a, b) \in (\mathbb{Z}^+)^2$ be a primitive, coprime pair of positive integer base coordinates. For any continuously varying real exponent parameter $t \in (0, \infty)$, the exact unregularized closure defect $D(t)$ relative to the classical Euclidean target baseline is defined as

$$D(t) = 1 - \frac{a^t + b^t}{(a^2 + b^2)^{t/2}}. \quad (1)$$

The structural behavior of this defect field is fundamentally governed by the continuous monotonicity of vector norms across mathematical spaces.

Theorem 1 (Positivity of the Integrated Closure Defect Functional). *Let $a, b \in \mathbb{Z}^+$ be positive coprime integer base coordinates. For any continuous deformation interval bounded by the target exponent $n > 2$, the Integrated Closure Defect Functional, defined as*

$$A_0(a, b; n) := \int_2^n D(t) dt, \quad (2)$$

is strictly positive:

$$A_0(a, b; n) > 0 \quad \forall n > 2. \quad (3)$$

Proof. Consider the classical mathematical theory of ℓ_p -spaces and the structural monotonicity of ℓ_p -norms. Related norm inequalities and asymptotic analytic techniques appear throughout

classical analysis and approximation theory [23, 27]. For any fixed vector with strictly positive real components (a, b) , the corresponding analytical norm

$$\|(a, b)\|_p = (a^p + b^p)^{1/p}$$

is a strictly decreasing continuous function with respect to the parameter p on the open interval $(0, \infty)$. Setting $p_1 = 2$ and $p_2 = t$, where $t > 2$, this monotonicity mandates that

$$(a^t + b^t)^{1/t} < (a^2 + b^2)^{1/2}.$$

Because both sides represent strictly positive real scalar quantities, raising this entire inequality to the power of t preserves the strict structural inequality

$$a^t + b^t < (a^2 + b^2)^{t/2}.$$

Dividing both sides by the right-hand geometric denominator yields

$$\frac{a^t + b^t}{(a^2 + b^2)^{t/2}} < 1.$$

Substituting this exact relation into the definition of the unregularized defect field established in Equation (1) proves that

$$D(t) > 0 \quad \forall t \in (2, n].$$

We note that this inequality holds universally for all positive coordinates. Even in the perfectly balanced isotropic state $a = b$, the functional does not collapse to zero for $t > 2$. Setting $a = b$ yields

$$D(t) = 1 - \frac{2a^t}{(2a^2)^{t/2}} = 1 - 2^{1-t/2}.$$

For all $t > 2$, $t/2 > 1$, implying $2^{1-t/2} < 1$, which preserves $D(t) > 0$. At the unique initialization boundary $t = 2$, the functional collapses identically to perfect classical Pythagorean closure:

$$D(2) = 1 - \frac{a^2 + b^2}{a^2 + b^2} = 0.$$

Since $D(t)$ is a continuous, non-negative real-valued function on the compact integration domain $[2, n]$ that is strictly positive on the non-degenerate open sub-interval $(2, n]$ for any $n > 2$, its standard Riemann integral across this parameter space must yield a strictly positive scalar output. Therefore,

$$A_0(a, b; n) = \int_2^n D(t) dt > 0.$$

This completes the formal proof of the positivity of the integrated closure defect functional.

Remark 1. *The proof remains valid for arbitrary positive real coordinates $(a, b) \in \mathbb{R}_{>0}^2$.*

□

3 Third-Order Asymptotic Expansion of A_0

To map the precise structural growth profile of this path-integrated field as it detaches from the classical baseline, we apply an asymptotic Maclaurin expansion directly to the interior integrand.

Let the distribution of the base components be parameterized via their normalized base weights, defined in APG I as

$$w_a = \frac{a^2}{a^2 + b^2}, \quad w_b = \frac{b^2}{a^2 + b^2}, \quad (4)$$

where $w_a + w_b = 1$. The structural information content of this coordinate configuration is measured in nats via its corresponding Shannon entropy:

$$H(W) = -(w_a \ln w_a + w_b \ln w_b). \quad (5)$$

The entropy functional originates in information theory [25, 6], while related divergence measures and information-theoretic variational principles were developed by Kullback and Leibler [19] and Jaynes [15].

For all positive base coordinates $a, b > 0$, the distribution satisfies $H(W) > 0$. If the system is perfectly balanced ($a = b$), the entropy reaches its global maximum value of $\ln 2$. We further isolate the quadratic logarithmic weight dispersion parameter $Q(a, b)$, defined as

$$Q(a, b) = w_a (\ln w_a)^2 + w_b (\ln w_b)^2. \quad (6)$$

The parameter $Q(a, b)$ measures the second-order logarithmic dispersion of the normalized weight distribution and naturally appears as the second-order coefficient in the local defect expansion. Note that $Q(a, b) > 0$ for all valid finite base coordinates since $0 < w_a, w_b < 1$, ensuring that $\ln w_a \neq 0$ and $\ln w_b \neq 0$ because neither weight can equal unity for positive finite a, b .

Theorem 2 (The Third-Order Asymptotic Expansion). *As the continuous target exponent moves in a local neighborhood of the classical Euclidean baseline ($n \rightarrow 2^+$), the Integrated Closure Defect Functional admits a third-order local asymptotic expansion:*

$$A_0(a, b; n) = \frac{H(W)}{4}(n-2)^2 - \frac{Q(a, b)}{24}(n-2)^3 + \mathcal{O}((n-2)^4). \quad (7)$$

Proof. We define the localized deformation variable $K_P = t - 2$, implying $t = 2 + K_P$. We expand the core constituent component of the numerator in Equation (1) utilizing the standard real analytic identity $x^y = \exp(y \ln x)$:

$$a^t = a^{2+K_P} = a^2 \exp(K_P \ln a).$$

Applying this exponential mapping to the normalized baseline fraction yields the exact structural transformation derived in APG I:

$$\frac{a^t}{(a^2 + b^2)^{t/2}} = \left[\frac{a^2}{a^2 + b^2} \right] \left[\frac{a}{\sqrt{a^2 + b^2}} \right]^{K_P} = w_a \exp\left(\frac{K_P}{2} \ln w_a\right).$$

Applying the canonical Taylor series expansion for the exponential function $\exp(x) = 1 + x + x^2/2 + \mathcal{O}(x^3)$ gives

$$w_a \exp\left(\frac{K_P}{2} \ln w_a\right) = w_a \left[1 + \frac{K_P}{2} \ln w_a + \frac{K_P^2}{8} (\ln w_a)^2 + \mathcal{O}(K_P^3) \right].$$

Replicating this exact algebraic sequence for the corresponding b -component and gathering the joint terms under the summation constraint $w_a + w_b = 1$ allows the exact defect field expression to be represented as

$$D(t) = 1 - \left[(w_a + w_b) + \frac{K_P}{2} (w_a \ln w_a + w_b \ln w_b) + \frac{K_P^2}{8} (w_a (\ln w_a)^2 + w_b (\ln w_b)^2) + \mathcal{O}(K_P^3) \right].$$

Canceling the leading constant unit terms and substituting the formal definitions of the Shannon entropy $H(W)$ and the dispersion factor $Q(a, b)$ establishes the local instantaneous defect expansion

$$D(t) = \frac{H(W)}{2} K_P - \frac{Q(a, b)}{8} K_P^2 + \mathcal{O}(K_P^3).$$

We evaluate the Integrated Closure Defect Functional by integrating this instantaneous expansion term by term along the continuous path parameter domain from $t = 2$ to $t = n$:

$$A_0(a, b; n) = \int_2^n D(t) dt = \int_0^{n-2} \left[\frac{H(W)}{2} K_P - \frac{Q(a, b)}{8} K_P^2 + \mathcal{O}(K_P^3) \right] dK_P.$$

Since the expansion is valid in a neighborhood of $K_P = 0$ and all terms are continuous, termwise integration preserves the asymptotic order. Evaluating the foundational primitive integral fields with respect to the deformation boundary gives

$$\int_0^{n-2} \frac{H(W)}{2} K_P dK_P = \left[\frac{H(W)}{4} K_P^2 \right]_0^{n-2} = \frac{H(W)}{4} (n-2)^2,$$

and

$$\int_0^{n-2} \frac{Q(a, b)}{8} K_P^2 dK_P = \left[\frac{Q(a, b)}{24} K_P^3 \right]_0^{n-2} = \frac{Q(a, b)}{24} (n-2)^3.$$

Integrating the higher-order remainder terms yields an integrated output order of $\mathcal{O}((n-2)^4)$. This completes the proof of the extended local asymptotic growth profile. $\square$

Equation (7) establishes that for any positive coordinate pair (a, b) , the initial detachment profile from the classical Euclidean target behaves as a local quadratic growth profile with cubic correction mediated jointly by the entropy and logarithmic weight dispersion variables.

4 The Conditional Frey-Curve Height Obstruction Program

Having established the internal mathematical laws governing the path-integrated functional A_0 , we introduce a conditional theoretical research program. This framework outlines the proposed cross-disciplinary criteria required to couple path-globalized continuous defect fields to the discrete arithmetic transformations of the modular landscape.

Let a hypothetical non-trivial primitive integer counterexample to Fermat's Last Theorem for an odd prime exponent $n > 2$ be encoded into the canonical Frey curve associated to a hypothetical primitive Fermat solution:

$$E_{a,b,n} : y^2 = x(x - a^n)(x + b^n). \tag{8}$$

The formal arithmetic invariants of this scheme, namely its minimal discriminant (Δ_E) and its conductor (N_E), are related to standard expressions up to powers of 2 and minimal-model normalization factors:

$$\Delta_E \asymp (abc)^{2n}, \quad N_E \text{ is controlled by } \text{rad}(abc). \quad (9)$$

We evaluate this system using the stable Faltings height $h_F(E)$, which scales the geometric volume of the minimal model.

Research Program: The Height-Strain Coupling Boundary

Under a generalized arithmetic framework, the stable Faltings height $h_F(E)$ is heuristically expected to grow comparably to $n \ln(abc)$, subject to minimal-model and archimedean correction terms. This statement is heuristic and is not used in any proof of the present paper. Because the components satisfy $abc \geq \text{rad}(abc)$, the prime objective of this conditional program is to establish whether the continuous integrated defect carries structural information bounding these arithmetic attributes. Similar questions concerning the interaction between arithmetic height functions, Diophantine complexity, and structural constraints arise throughout modern arithmetic geometry and Diophantine approximation theory [30, 5, 16, 17].

Classical proofs of Fermat’s Last Theorem derive a structural contradiction through the incompatibility between the Frey curve and the modular forms predicted by Galois representation level-lowering. To connect APG to this modular obstruction field, the framework must progress from a descriptive dictionary to an explicit arithmetic inequality. The proposed bridge theorem of this program is to rigorously investigate whether the continuous path integral forces a lower boundary of the form

$$h_F(E_{a,b,n}) \geq C A_0(a, b; n) \quad \text{or} \quad \ln |\Delta_E| \geq C A_0(a, b; n) \quad (10)$$

for a positive arithmetic scalar C . Proving an inequality of this type requires demonstrating that the continuous path integral isolates an unclosable geometric defect field that cannot be accommodated within the discrete conductor bounds permitted by a stable modular representation—an open cross-disciplinary path that remains a subject for future research.

5 Asymptotic Invariant Dynamics

To illustrate the asymptotic behavior of the unregularized path mechanics, the exact Integrated Closure Defect Functional was computed across a representative integer coordinate landscape using a standard numerical quadrature routine. Table 1 reports the numerical values of the exact continuous defect path integral (A_0) against the extended third-order asymptotic expansion derived in Theorem 2 for the canonical coprime base coordinate framework $(a, b) = (3, 4)$.

$$\begin{aligned} \text{Given: } a = 3, \quad b = 4, \quad c_2 = 5, \quad w_a = 0.36, \quad w_b = 0.64, \\ H(W) = 0.653418 \text{ nats}, \quad Q(a, b) = w_a (\ln w_a)^2 + w_b (\ln w_b)^2 = 0.505703. \end{aligned}$$

The numerical data reported above illustrate the asymptotic behavior modeled in the text. Near the classical baseline ($n \leq 2.10$), the close alignment numerically illustrates that the initial rate of global defect accumulation is governed to leading order by the quadratic entropy expansion. Values for $n = 3$ and $n = 4$ are explicitly included only to illustrate the expected deterioration of a local asymptotic approximation away from its original expansion point. As the parameter advances into higher regimes ($n \geq 3$), the higher-order terms become visible, and the growth curves diverge predictably, isolating the structural contributions modeled by the $\mathcal{O}((n-2)^4)$ remainder envelope.

Table 1: Numerical Tracking of Integrated Closure Defect Invariants

Exponent (n)	Parameter ($n - 2$)	Exact Integral (A_0)	Asymptotic Approx. (Eq. 7)	Absolute Error
2.00	0.00	0.000000	0.000000	0.000000
2.01	0.01	0.000016	0.000016	0.000000
2.02	0.02	0.000065	0.000065	0.000000
2.05	0.05	0.000405	0.000406	0.000001
2.10	0.10	0.001611	0.001613	0.000002
2.20	0.20	0.006361	0.006365	0.000004
3.00	1.00	0.133211	0.142284	0.009073
4.00	2.00	0.495034	0.611511	0.116477

6 Conclusion and Strategic Status

Arithmetic Power Geometry III completes a major internal evolution by upgrading the framework from a local analytical dictionary into an integrated path-analysis system. By proving the positivity (Theorem 1) and the third-order local asymptotic expansion (Theorem 2) of the unregularized integrated closure defect functional A_0 , the internal consistency of the continuous deformation landscape is rigorously secured.

Crucially, the framework does not claim an independent or completed proof of Fermat’s Last Theorem, the *abc* conjecture, or a finalized Frey-Faltings height obstruction boundary. The rigorous results of this manuscript are strictly restricted to the proven analytical properties of the internal defect invariant A_0 . The proposed interface with the Faltings height and modular constraints is explicitly positioned as a conditional research program. Future work within this discipline will focus on analyzing the bounding alignment of minimal models to determine if a formal arithmetic bridge theorem can be achieved.

References

- [1] Akhtar, M. A. K. (2026a). *Foundations of Arithmetic Power Geometry: Local Parameter Deformations and Entropy at the Euclidean Target*. Preprint.
- [2] Akhtar, M. A. K. (2026b). *Foundations of Arithmetic Power Geometry II: Information-Geometric Formulations, Coordinate Stability, and Logarithmic Projective Height Mappings*. Preprint.
- [3] Amari, S. I. (2016). *Information Geometry and Its Applications*. Springer.
- [4] Amari, S. I., & Nagaoka, H. (2000). *Methods of Information Geometry*. American Mathematical Society.
- [5] Bombieri, E., & Gubler, W. (2006). *Heights in Diophantine Geometry*. Cambridge University Press.
- [6] Cover, T. M., & Thomas, J. A. (2006). *Elements of Information Theory* (2nd ed.). Wiley-Interscience.
- [7] Bugeaud, Y. (2004). *Approximation by Algebraic Numbers*. Cambridge University Press.

- [8] Cassels, J. W. S. (1957). *An Introduction to Diophantine Approximation*. Cambridge University Press.
- [9] Darmon, H., & Granville, A. (1995). On the equations $z^m = F(x, y)$ and $ax^p + by^q = cz^r$. *Bulletin of the London Mathematical Society*, 27(6), 513–543.
- [10] Davis, P. J., & Rabinowitz, P. (2007). *Methods of Numerical Integration*. Courier Corporation.
- [11] Do Carmo, M. P. (1992). *Riemannian Geometry*. Birkhäuser.
- [12] Faltings, G. (1983). Endlichkeitssätze für abelsche Varietäten über Zahlkörpern. *Inventiones Mathematicae*, 73(3), 349–366.
- [13] Frey, G. (1986). Links between stable elliptic curves and certain Diophantine equations. *Annales Universitatis Saraviensis*, 1(1), 1–40.
- [14] Hartshorne, R. (1977). *Algebraic Geometry*. Springer-Verlag.
- [15] Jaynes, E. T. (1957). Information theory and statistical mechanics. *Physical Review*, 106(4), 620–630.
- [16] Hindry, M., & Silverman, J. H. (2000). *Diophantine Geometry: An Introduction*. Springer-Verlag.
- [17] Lang, S. (2002). *Algebra* (3rd ed.). Springer-Verlag.
- [18] Lee, J. M. (2018). *Introduction to Riemannian Manifolds* (2nd ed.). Springer.
- [19] Kullback, S., & Leibler, R. A. (1951). On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1), 79–86.
- [20] Masser, D. W. (1985). Open problems. In *Proceedings of the Symposium on Analytic Number Theory*. Imperial College.
- [21] Ribet, K. A. (1990). On modular representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ arising from modular forms. *Inventiones Mathematicae*, 100(1), 431–476.
- [22] Oesterlé, J. (1988). Nouvelles approches du “théorème” de Fermat. *Séminaire Bourbaki*, 30, 165–186.
- [23] Rudin, W. (1976). *Principles of Mathematical Analysis* (3rd ed.). McGraw-Hill.
- [24] Schmidt, W. M. (1980). *Diophantine Approximation*. Springer-Verlag.
- [25] Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423.
- [26] Stewart, C. L., & Tijdeman, R. (1986). On the Oesterlé-Masser conjecture. *Monatshefte für Mathematik*, 102(3), 251–257.
- [27] Whittaker, E. T., & Watson, G. N. (1927). *A Course of Modern Analysis* (4th ed.). Cambridge University Press.
- [28] Silverman, J. H. (2007). *The Arithmetic of Elliptic Curves* (2nd ed.). Springer.

- [29] Taylor, R., & Wiles, A. (1995). Ring-theoretic properties of certain Hecke algebras. *Annals of Mathematics*, 141(3), 553–572.
- [30] Vojta, P. (1987). *Diophantine Approximations and Value Distribution Theory*. Springer-Verlag.
- [31] Wiles, A. (1995). Modular elliptic curves and Fermat’s Last Theorem. *Annals of Mathematics*, 141(3), 443–551.